STANDARD DEVIATION

A Step-by-Step Beginner's Guide with Python and NumPy

1. What Is Standard Deviation?

Standard deviation (SD) measures how spread out values are around their mean (average). A small SD means values tend to stay close to the mean; a large SD means values tend to be farther away.

Example: If two groups both have an average score of 50, but one group's scores are mostly 48–52 and another group's scores range widely, the second group has a larger standard deviation.

2. Why Do We Use It?

• **Compare consistency:** See whether students' marks or product measurements are tightly grouped or widely spread.

• **Understand data variation:** The mean alone does not tell you whether values are close together or far apart.

• **Interpret random data:** In a normal distribution, SD describes the typical scale of spread around the mean.

• **Spot unusual values:** SD can help identify values far from the mean, but it does not automatically prove that a value is an error.

3. Worked Example: Calculate SD by Hand

Use these five marks: 40, 45, 50, 55, 60. We will calculate the **population SD**, dividing by N = 5.

Step 1 — Calculate the mean

Mean = (40 + 45 + 50 + 55 + 60) / 5 = **50**

Value (x)	x – Mean	(x – Mean)²
40	40 – 50 = –10	100
45	45 – 50 = –5	25
50	50 – 50 = 0	0
55	55 – 50 = 5	25
60	60 – 50 = 10	100
Total	**0**	**250**

Step 2 — Subtract the mean from every value. These differences are called deviations.

Step 3 — Square every deviation. Squaring removes negative signs and gives larger deviations more weight.

Step 4 — Calculate the variance

Variance = Sum of squared deviations / N = 250 / 5 = **50**

Step 5 — Take the square root

SD = √Variance = √50 ≈ **7.07**

Interpretation: the values have a standard deviation of about 7.07 marks around a mean of 50. SD is a measure of spread, not a fixed gap between consecutive values.

4. The Formula

Population standard deviation: $\sigma = \sqrt{[\ \Sigma(x - \mu)^2 / N\ ]}$

• x = each data value; μ = population mean; N = number of values; Σ means “sum them all.”

Sample standard deviation: $s = \sqrt{[\ \Sigma(x - x\blacksquare)^2 / (n - 1)\ ]}$

Use the population formula when describing the entire population. Use the sample formula when estimating spread from a sample. NumPy’s np.std(data) uses the population version by default (ddof=0). Use np.std(data, ddof=1) for the sample version.

5. Calculate It in Python

```
import numpy as np

data = np.array([40, 45, 50, 55, 60])

mean = np.mean(data)
deviations = data - mean
squared_deviations = deviations ** 2
variance = np.mean(squared_deviations)
sd_manual = np.sqrt(variance)

print("Data:", data)
print("Mean:", mean)
print("Deviations:", deviations)
print("Squared deviations:", squared_deviations)
print("Variance:", variance)
print("Manual SD:", sd_manual)
print("NumPy SD:", np.std(data))
```

Expected key results

Quantity	Result
Mean	50.0
Deviations	[-10, -5, 0, 5, 10]
Squared deviations	[100, 25, 0, 25, 100]
Variance	50.0
Standard deviation	7.0710678…

6. Standard Deviation in np.random.normal()

Syntax: np.random.normal(loc, scale, size)

• loc = requested mean (center), scale = requested standard deviation (spread), size = number of values generated.

```
np.random.seed(42)

low_spread = np.random.normal(50, 2, 10)
high_spread = np.random.normal(50, 15, 10)

print("Low-spread data:", low_spread)
print("Actual SD:", np.std(low_spread))

print("High-spread data:", high_spread)
print("Actual SD:", np.std(high_spread))
```

Important: In np.random.normal(50, 15, 10), 15 is the distribution's requested standard deviation, not a fixed difference between adjacent values. The actual SD of just 10 generated values will usually be near, but not exactly, 15. With more generated values, the measured SD tends to be closer to the requested scale.

7. What Does SD Tell Us in a Normal Distribution?

For an ideal normal distribution, approximately:

• 68% of values lie within 1 SD of the mean (mean ± 1 SD).

• 95% lie within 2 SDs of the mean.

• 99.7% lie within 3 SDs of the mean.

These are approximate proportions for the distribution, not guarantees for every small random sample.

8. Common Mistakes to Avoid

• Do not treat standard deviation as the fixed gap between values.

• Do not call np.random.normal() repeatedly inside one calculation if you intend to analyze one dataset. Save the generated values in a variable such as data.

• Do not use np.sqrt(data - mean) as the SD formula. Square the deviations first, average the squared deviations, then take the square root.

• The seed makes a pseudo-random sequence reproducible when the same setup is rerun; it does not set the standard deviation or control the gaps.

9. Quick Summary

Mean tells you the center. **Standard deviation** tells you the spread. **Variance** is the average squared deviation (for the population formula). In NumPy, **np.std(data)** calculates population SD by default, while **np.std(data, ddof=1)** calculates sample SD.

Practice question

For data = [10, 10, 10, 10], what are the mean and standard deviation? Think about how spread out the values are.